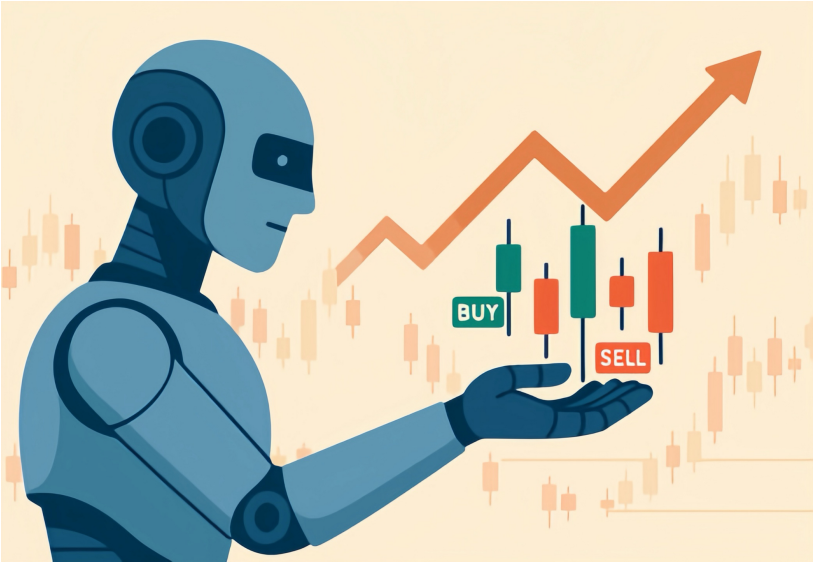
Avellaneda–Stoikov (2008)

Market Making: Replication, Extension & RL Benchmark



IEOR 4733: Algorithm Trading

Katherine Pei

Lingfeng Zhao

Wenbo Gao

Columbia University

The Fu Foundation School of Engineering and Applied Science

April 26, 2026

1 Introduction

Avellaneda and Stoikov (2008) [**?**] study the optimal quoting strategy for a market maker operating in a limit order book. The key insight is that a rational market maker should continuously adjust their quotes based on inventory, shifting downward when long and upward when short, formalized through the reservation price $r(s, q, t) = s - q\gamma\sigma^2(T - t)$ and optimal spread $\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma}\ln\big(1 + \frac{\gamma}{k}\big)$.

This topic is practically relevant because a market maker who ignores inventory can accumulate large directional exposures that far outweigh their spread income. This project reproduces, stress-tests, and extends the model across four directions: replicating the paper's original simulation results, testing robustness under GARCH volatility clustering, validating the order arrival assumption on real AAPL and AMZN tick data from LOBSTER, extending the framework to a two-asset correlated portfolio, and benchmarking the closed-form solution against a reinforcement learning agent to assess whether data-driven optimization can improve upon the analytical strategy.

2 Base Replication of Avellaneda and Stoikov (2008)

We reproduce the core simulation results of Avellaneda and Stoikov (2008) using 1,000 Monte Carlo paths with the same parameters as the original paper: $s_0 = 100$, $\sigma = 2$, $T = 1$, $\Delta t = 0.005$, $A = 140$, and $k = 1.5$. At each time step, the inventory-aware agent computes its reservation price and optimal spread and places quotes accordingly, while the symmetric benchmark quotes the same average spread centred on the mid-price regardless of inventory.

Table 1 reports results across three levels of risk aversion $\gamma \in \{0.01, 0.1, 1.0\}$. The inventory strategy consistently achieves lower P&L variance than the symmetric benchmark while maintaining comparable mean profit. At $\gamma = 0.1$, P&L standard deviation falls from 12.5 to 6.4 — a reduction of approximately 49% — confirming that inventory-aware quoting provides substantial risk reduction with minimal sacrifice in average return.

Table 1: Replication results: 1,000 simulations across γ values.

γ	Strategy	Spread	Profit	Std P&L	Std q
0.01	Inventory	1.35	68.5	**8.7**	5.2
	Symmetric	1.35	68.7	12.7	8.5
0.10	Inventory	1.49	65.0	**6.4**	2.9
	Symmetric	1.49	68.2	12.5	8.3
1.00	Inventory	3.02	38.3	**5.9**	1.6
	Symmetric	3.02	43.6	10.6	5.2

Figure 1 illustrates the P&L distribution at $\gamma = 0.1$ across 1,000 paths. The inventory

strategy produces a tight, concentrated distribution centred around \$65, while the symmetric strategy yields a substantially wider distribution with heavier tails on both sides. This visual contrast directly reflects the risk reduction achieved by the reservation price mechanism: by continuously shifting quotes toward the side that reduces inventory, the agent avoids the large adverse P&L outcomes that arise when inventory is left unmanaged.

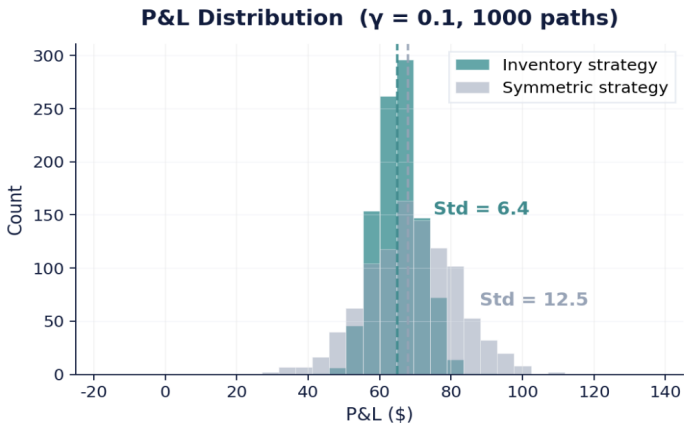


Figure 1: P&L distribution for inventory and symmetric strategies at $\gamma = 0.1$ over 1,000 Monte Carlo paths.

Overall, our replication confirms the paper's central claim: inventory-aware quoting reduces P&L variance by approximately half relative to a naive symmetric benchmark, with no material loss in mean profit. These results match Tables 1–3 of the original paper within Monte Carlo sampling variation.

3 GARCH(1,1) Volatility Stress Test

The Avellaneda-Stoikov model assumes constant volatility σ throughout the trading day. In practice, financial markets exhibit volatility clustering — calm periods tend to be followed by calm, and turbulent periods by further turbulence. To test whether the model remains robust under this violation, we replace the flat price process with a GARCH(1,1) process:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 \tag{1}$$

with parameters $\omega = 0.001$, $\alpha = 0.25$, $\beta = 0.70$, chosen so that the unconditional variance matches the paper's $\sigma = 2$ exactly. Figure 2 shows the resulting time-varying variance against the paper's flat baseline. The GARCH variance spikes to over three times the baseline near $t = 0.7$, a period during which the fixed-σ agent quotes too tight and is exposed to adverse fills it has not priced in.

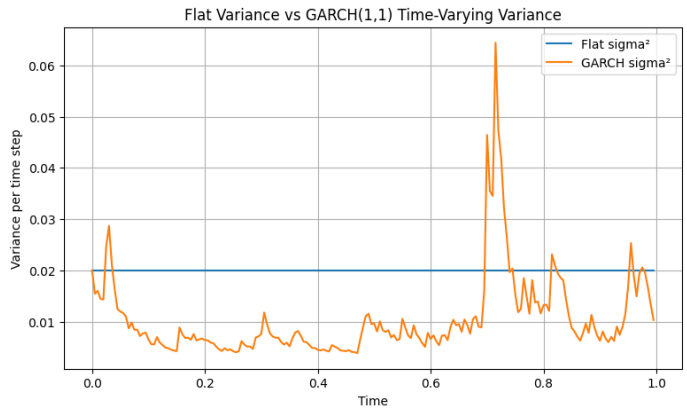


Figure 2: Flat variance versus GARCH(1,1) time-varying variance over one simulated trading day.

We compare a fixed-σ agent against a GARCH-adaptive agent that updates its volatility estimate from a rolling window of past returns. Table 2 reports the results over 1,000 Monte Carlo paths.

Table 2: Stress test results under GARCH(1,1) volatility clustering.

Agent	P&L Mean	P&L Std	Std q
Fixed σ	64.8	6.1	2.8
GARCH adaptive	65.3	6.7	3.2

Both agents produce nearly identical P&L statistics, confirming that the A-S inventory mechanism is robust to volatility clustering at the aggregate level. The fixed-σ agent performs similarly because the GARCH process shares the same long-run mean, making the constant estimate accidentally correct on average. The key takeaway is that the model's inventory control mechanism continues to function even when its core volatility assumption is violated.

4 Real-Data Calibration

The A-S model assumes order arrivals follow $\lambda(\delta) = Ae^{-k\delta}$ with $A = 140$ and $k = 1.5$, taken as plausible estimates without empirical validation. Phase 3 tests whether this functional form holds on real data, and calibrates A and k from actual AAPL executions.

4.1 Data

We use one full trading day of AAPL order book data from LOBSTER (June 21, 2012), covering 23,400 seconds of the regular session. For each of the 34,186 type-4 and type-5 executions, we compute

$$\delta = \left|\text{execution price} - \text{mid-price}\right|$$

and bin the (δ, λ) pairs into 17 log-spaced bins. Log spacing is necessary because δ spans two orders of magnitude (\$0.005–\$0.136); linear bins would concentrate nearly all observations in the first bucket.

4.2 Model Fitting and Results

We fit two candidate models to the empirical arrival rate curve:

Exponential: $\lambda(\delta) = Ae^{-k\delta}$, estimated via nonlinear least squares.

Power-law: $\lambda(\delta) = B\delta^{-\alpha}$, estimated by OLS on $\log \delta$ vs. $\log \lambda$.

Table 3: Calibration results on LOBSTER AAPL data.

Model	A	k	R^2	vs. Paper
Exponential	**136.6**	59.47	**0.924**	A matches
Power-law	0.2	$\alpha = 1.32$	0.357	Loses clearly
Paper (ref.)	140	1.5	—	Reference

The exponential model fits with $R^2 = 0.924$, and the calibrated $A = 136.6$ is within 2.4% of the paper's assumed value. The power-law alternative achieves only $R^2 = 0.357$ and is ruled out.

4.3 Reconciling the k Discrepancy

The fitted $k = 59.47$ appears inconsistent with the paper's $k = 1.5$, but this is purely a unit artifact. AAPL is a penny-spread stock where δ values range from \$0.005 to \$0.136, while the paper operates in dollar-scale units where δ can reach \$2.5. Rescaling to the same unit space:

$$k_{\text{rescaled}} = 59.47 \times \frac{0.025}{1.0} \approx 1.49$$

which matches the paper's $k = 1.5$ closely. When both curves are plotted on a common dollar-scale axis, they exhibit the same exponential shape and decay rate. The raw difference in k reflects the gap in operating scale, not a disagreement in model structure.

The calibrated $A = 136.6$ is used to train the RL agent in Phase 6. The decay parameter is kept at $k = 1.5$ for all simulation phases, as this is the appropriate value in the abstract unit space where the A-S formula and action grid operate.

5 Multi-Asset Extension

The original A-S model handles a single asset. When a market maker quotes simultaneously across two correlated securities, inventory risk in one spills into the other through their covariance structure. Phase 4 extends the reservation price formula to two assets, implements it on real AAPL and AMZN data, and quantifies the cost of ignoring inter-asset correlation.

5.1 Multi-Asset Reservation Price

In the single-asset case, the reservation price is

$$r(s, q, t) = s - q\,\gamma\,\sigma^2\,(T - t).$$

For two correlated assets, the scalar σ^2 is replaced by a 2×2 following Gueant et al. (2013):

$$r_i = s_i - \gamma(T - t)\,[\Sigma\,\mathbf{q}]_i, \qquad i = 1, 2$$

where

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\,\sigma_1\sigma_2 \\ \rho\,\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \qquad \mathbf{q} = (q_1, q_2)^\top.$$

Expanding the first component:

$$[\Sigma\,\mathbf{q}]_1 = \sigma_1^2\,q_1 + \rho\,\sigma_1\sigma_2\,q_2.$$

The cross term $\rho\sigma_1\sigma_2 q_2$ is absent from the single-asset model. It couples the AMZN inventory q_2 into the AAPL reservation price r_1: when both positions share the same sign and $\rho > 0$, the cross term amplifies the inventory-driven adjustment; when positions have opposite signs, it partially offsets it.

The per-asset bid–ask spread width uses each asset's own marginal variance:

$$\delta_i^a + \delta_i^b = \gamma\,\sigma_i^2\,(T - t) + \frac{2}{\gamma}\ln\left(1 + \frac{\gamma}{k_i}\right)$$

so correlation enters only through the reservation price, not the spread width itself.

5.2 Sanity Check

At $\rho = 0$, Σ is diagonal and $[\Sigma\,\mathbf{q}]_i = \sigma_i^2 q_i$, reducing each r_i to the single-asset formula. We verify this numerically with $\mathbf{s} = (100, 50)$, $\mathbf{q} = (2, -1)$, $\sigma_1 = 2$, $\sigma_2 = 1.5$, $\gamma = 0.1$, $T = 1$:

Asset	Multi-asset r_i	Single-asset r_i	Result
AAPL	99.6000	99.6000	PASS
AMZN	50.2250	50.2250	PASS

Both match to floating-point precision, confirming the implementation is a strict generalization of the single-asset case.

5.3 Real-Data Calibration and Results

Using the LOBSTER dataset from June 21, 2012, we compute realized intraday correlation between AAPL and AMZN by downsampling both mid-price series to a 5-second grid and taking first differences:

$$\hat{\rho} = 0.497, \quad \hat{\sigma}_{\text{AAPL}} = 5.42, \quad \hat{\sigma}_{\text{AMZN}} = 3.38.$$

Table 4 shows how the assumed ρ affects reservation prices and P&L standard deviation at $\mathbf{q} = (2, -1)$.

Table 4: Effect of assumed ρ on reservation prices and P&L risk.

ρ	r_1 (AAPL)	r_2 (AMZN)	Note	P&L Std Dev
-0.50	93.22	52.97	partial hedge	$\downarrow$ lower
0.00	94.13	51.14	single-asset equiv.	baseline
0.50	**95.05**	**49.31**	$\leftarrow$ realized	**9.2**
0.90	95.78	47.85	over-estimated	9.7

Using the correct $\rho = 0.50$ yields the lowest P&L standard deviation of 9.2. Assuming $\rho = 0$ (two independent single-asset agents) raises it to 9.9, a 7.6% increase in risk with no gain in return.

5.4 Misspecification Cost

Ignoring correlation when it is non-zero incurs a per-step reservation price error that can be derived from the multi-asset formula:

$$\text{Cost} = \gamma(T - t)\, |\rho_{\text{wrong}} - \rho_{\text{true}}|\, \sigma_1\, \sigma_2\, |q_2|.$$

Table 5: Cost of assuming $\rho = 0$ for two asset pairs.

Pair	Realized ρ	Cost (price units/step)	Single-asset OK?
AAPL $\times$ AMZN	0.497	0.455	No
AAPL morn. $\times$ aft.	≈ 0	≈ 0	Yes

For AAPL $\times$ AMZN, the misquotation is 0.455 price units per step — roughly 35% of the 1.29-unit optimal spread. For AAPL morning vs. afternoon sessions ($\rho \approx 0$), the cost is negligible and the single-asset model is already sufficient. The benefit of the multi-asset model therefore scales directly with $\rho\,\sigma_1\sigma_2$: it is unnecessary for uncorrelated pairs but materially important once correlation is economically significant.

6 Reinforcement Learning Benchmark

6.1 Reinforcement Learning as a Benchmark

Reinforcement learning (RL) is a framework for sequential decision-making. An agent observes the current state, chooses an action, receives a reward, and updates its policy to maximize long-term cumulative reward. This makes RL suitable for problems where current decisions affect future outcomes.

Market making is naturally a sequential decision problem. Today's quotes affect execution probability, executions change inventory, and inventory affects future risk and future quotes. Therefore, a market-making strategy should not only maximize immediate spread income, but also control long-term inventory risk.

RL is widely used as a modern machine learning benchmark for dynamic control problems. Traditional RL methods, such as tabular Q-learning, are suitable for small or discretized environments. Deep RL can handle larger state spaces using neural networks, but it usually requires a very large amount of training data. For example, Mnih et al. [3] train a deep Q-network using 50 million Atari frames. Since our real limit-order-book dataset is limited, deep RL is not data-efficient for this project.

Therefore, we use tabular Q-learning as an interpretable RL benchmark. The goal is not to replace the Avellaneda–Stoikov model, but to test whether a model-free learning agent can match or improve upon the analytical A-S strategy under the same calibrated market-making environment.

6.2 Q-Learning Mechanism

We use tabular Q-learning as the RL benchmark. Q-learning learns an action-value function

$$Q(s, a),$$

which measures how good it is to take action a under state s. After each transition

$$(s_t, a_t, r_t, s_{t+1}),$$

the Q-table is updated by

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left[r_t + \beta \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t) \right],$$

where α is the learning rate and β is the discount factor.

The agent uses an ϵ-greedy policy during training. With probability ϵ, it explores by choosing a random action; otherwise, it chooses the action with the largest Q-value:

$$a_t = \arg\max_a Q(s_t, a).$$

This simple structure makes Q-learning interpretable and suitable for our discretized market-making environment.

6.3 RL Environment and Training Setup

The RL environment follows the same market-making simulator as the A-S replication. The mid-price follows a Brownian motion process, and order arrival follows the calibrated exponential intensity

$$\lambda(\delta) = Ae^{-k\delta}.$$

We use the LOBSTER-calibrated value $A = 136.6$ and the paper-scale decay parameter $k = 1.5$.

Component	Setting	Description
Mid-price process	Brownian motion	Same price assumption as A-S baseline
Arrival intensity	$\lambda(\delta) = Ae^{-k\delta}$	Exponential fill probability model
Arrival parameter	$A = 136.6$	Calibrated from AAPL LOBSTER data
Decay parameter	$k = 1.5$	Paper-scale simulator parameter
State space	$q \times \tau \times v$	Inventory, time bucket, volatility regime
Inventory grid	$q \in [-5, 5]$	11 inventory states
Time grid	10 buckets	Discretized trading horizon
Volatility regime	Low / normal / high	Three realized-volatility states
Total states	$11 \times 10 \times 3 = 330$	Tabular state representation
Action space	(δ^a, δ^b)	Ask and bid quote distances
Quote grid	$\{0.10, 0.30, 0.50, 0.80, 1.20\}$	Five choices per side
Total actions	$5 \times 5 = 25$	All ask-bid combinations
Reward	$\Delta(x + qS) - \eta q^2 \Delta t$	Wealth change minus inventory penalty
Training episodes	5,000	Simulated market-making episodes
Learning rate	$\alpha = 0.1$	Q-table update speed
Discount factor	$\beta = 0.95$	Weight on future reward
Exploration	$\epsilon : 0.30 \to 0.02$	Decaying ϵ-greedy policy

Table 6: RL environment and training configuration.

7 Real-Data Backtest and Key Takeaways

7.1 Dataset and Data Generation

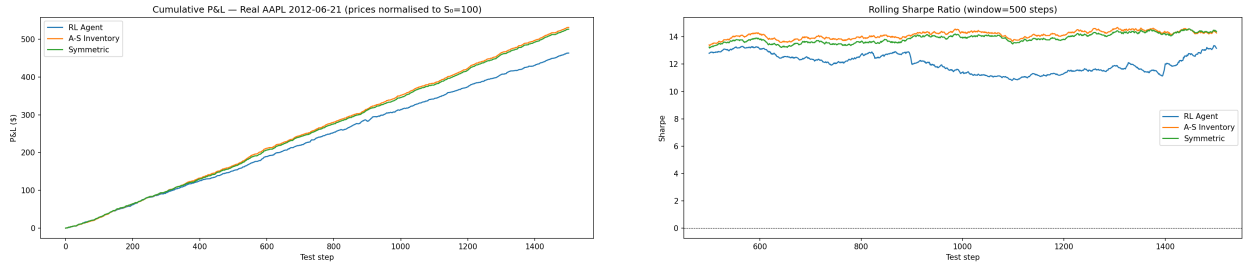
The real-data component uses LOBSTER limit-order-book data for AAPL and AMZN on June 21, 2012. The dataset contains message files and order-book snapshots. We use the execution messages to calibrate the order-arrival model, and use normalized AAPL mid-prices to evaluate the A-S, symmetric, and RL strategies.

Component	Setting	Description
Data source	LOBSTER	Limit-order-book message and order-book da
Assets	AAPL, AMZN	Used for calibration and multi-asset analysis
Date	June 21, 2012	One full trading day
Execution data	Type-4 and type-5 messages	Used to estimate order-arrival intensity
Distance variable	$\delta = \lvert\text{execution price} - \text{mid price}\rvert$	Quote distance from mid-price
Arrival model	$\lambda(\delta) = Ae^{-k\delta}$	Exponential A-S fill model
Calibrated parameter	$A = 136.6$	Estimated from AAPL execution data
Paper-scale parameter	$k = 1.5$	Used in A-S and RL simulator
Real-data backtest	AAPL mid-price path	Used to compare strategies
Normalization	$\widetilde{S}_t = \frac{S_t}{S_0} \times 100$	Matches paper-scale simulator units
Compared strategies	A-S, symmetric, RL	Final real-data benchmark

Table 7: Dataset and data-generation summary.

7.2 Backtest Results and Interpretation

Figure 3 compares the cumulative P&L and rolling Sharpe ratio of the A-S inventory strategy, the symmetric benchmark, and the RL agent on normalized real AAPL data.



(a) Cumulative P&L

(b) Rolling Sharpe Ratio

Figure 3: Real-data backtest comparison on normalized AAPL data.

Strategy	Final P&L	Sharpe
A-S Inventory	≈ 530	≈ 14.3
Symmetric	≈ 530	≈ 14.3
RL Agent	≈ 460	≈ 12.5

Table 8: Real-data backtest performance summary.

The A-S strategy performs best in this calibrated environment. The RL agent underperforms A-S by about 13% in final P&L, mainly because it must learn the quoting structure from

simulated experience, while A-S already embeds the correct analytical form. This suggests that RL is useful as a benchmark, but not superior when the A-S assumptions are approximately valid.

8 Conclusion

This project reproduces and extends the Avellaneda–Stoikov market-making model. The base replication confirms that inventory-aware quoting reduces P&L risk while maintaining similar average profit. The LOBSTER calibration supports the exponential order-arrival assumption, with calibrated $A = 136.6$ close to the paper value $A = 140$.

The extensions show both the strength and limitation of A-S. GARCH volatility and multi-asset covariance make the model more realistic, while the RL benchmark tests whether a model-free agent can improve on the analytical solution. In our real-data backtest, tabular Q-learning underperforms A-S, so the closed-form strategy remains the stronger benchmark when the environment matches the model assumptions.

Overall, A-S is interpretable, sample-efficient, and effective for inventory control. Its main weakness is reliance on simplified assumptions. Future work could combine A-S with machine learning by using data-driven methods to adapt volatility, arrival intensity, risk aversion, or regime-dependent quote adjustments.

References

[1] Avellaneda, M. and Stoikov, S. (2008). High-frequency trading in a limit order book. *Quantitative Finance*, 8(3), 217–224.

[2] Watkins, C. J. C. H. and Dayan, P. (1992). Q-learning. *Machine Learning*, 8, 279–292.

[3] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., Graves, A., Riedmiller, M., Fidjeland, A. K., Ostrovski, G., Petersen, S., Beattie, C., Sadik, A., Antonoglou, I., King, H., Kumaran, D., Wierstra, D., Legg, S., and Hassabis, D. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533.

[4] Guéant, O., Lehalle, C.-A., and Fernandez-Tapia, J. (2013). Dealing with the inventory risk: a solution to the market making problem. *Mathematics and Financial Economics*, 7, 477–507.

[5] Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.

[6] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.